BERT vs GPT CHEATSHEET

AI Engineer Ready · Week 2 — Encoder-Only vs Decoder-Only Models

Golden Rule → need understanding/classification? BERT. need generation? GPT.

1. Side-by-Side

	BERT	GPT
Architecture	Encoder-only	Decoder-only
Reads text	Bidirectionally (both directions at once)	Left-to-right (autoregressive)
Trained with	Masked Language Modeling (predict a hidden [MASK] token)	Next-token prediction
Best at	Understanding, classification, embeddings	Open-ended generation, chat, reasoning
Typical use	Sentiment analysis, search relevance, NER	Chatbots, drafting, code generation

2. BERT — Masked Language Modeling

```
$ "The [MASK] sat on the mat." -> BERT predicts: "cat" (uses BOTH left and right context)
```

BERT sees the whole sentence at once (minus the masked token), so it can use context from **both** before and after the blank — that's what "bidirectional" means.

3. Context Sensitivity (Same [MASK], Different Meaning)

Sentence	BERT's top prediction for [MASK]
"I went to the bank to deposit money."	financial institution sense of "bank"
"I sat by the river bank."	riverside sense of "bank"

Same masked word, different surrounding context → different prediction. This is why contextual embeddings (from models like BERT) beat static word vectors: the representation of "bank" changes depending on the sentence.

4. GPT — Decoding Settings

Setting	Low value	High value
Temperature	More deterministic, focused, repetitive	More random, creative, varied
Typical use	Factual Q&A, code, extraction	Brainstorming, creative writing

5. Sentence-Level Similarity with BERT

```
$ from transformers import AutoTokenizer, AutoModel
$ tok = AutoTokenizer.from_pretrained("bert-base-uncased")
$ model = AutoModel.from_pretrained("bert-base-uncased")
$ # use the [CLS] token's embedding to represent the whole sentence,
```

```
$ # then compare sentences with cosine similarity
```

Quick decision rule

If you need a label, a score, or a search-ranking embedding -> reach for an encoder (BERT-family). If you need free-form text generated for a user -> reach for a decoder (GPT-family).